

Quiz

- Due No due date
- Points 20
- Questions 19
- Available Feb 21, 2024 at 11:40am - Feb 21, 2024 at 11:55am 15 minutes
- Time Limit None

Instructions

- Rules for taking the exam:
 - You should only use one browser window that you use to take the Quiz. No other windows/tabs or other applications should be open on your computer.
 - You must complete the midterm on your own. You are not allowed to collaborate with anyone else, regardless of whether they are in the class or not.
 - This is a closed-book exam and as such you are not allowed to consult any of the course materials such as lecture notes, homework solutions, lab slides, or the textbook, as well as any online help.

This quiz was locked Feb 21, 2024 at 11:55am.

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	14 minutes	19.97 out of 20

❗ Correct answers are no longer available.

Score for this quiz: 19.97 out of 20

Submitted Feb 21, 2024 at 11:54am

This attempt took 14 minutes.



PartialQuestion 1

0.5 / 1 pts

Select all the data structures that maintain the order of elements in Python

☒ [1, 4, 7]

☐ {56, 78, 90}

☒ {'a':1, 'b':2}

☒ (4, 6, 9)



Question 2

1 / 1 pts

Select the data structure which does not allow duplicate values

☐ List

☒ Set

☐ Tuple



Question 3

1 / 1 pts

Select the immutable data structures

☒ String

☒ Tuple

☐ List

☐ Dictionary



Question 4

1 / 1 pts

Which of the below built-in function will you use to find the data type of a variable 'a'

☐ instance(a)

☒ type(a)

☐ typeof(a)



Question 5

1 / 1 pts

If you want your function named 'doSomething' to receive a variable list of arguments, then the first line of the function definition should be

☒ def doSomething(*a):

☐ def doSomething(**a):

☐ def doSomething({a}):



Question 6

1 / 1 pts

Which SQL keyword will you use to get unique values in a column?

☐ UNIQUE

☒ DISTINCT

☐ DIFFERENT



Question 7

1 / 1 pts

Which SQL operator is used to select values within a range that is used in a WHERE clause?

☒ BETWEEN

☐ WITHIN

☐ RANGE



Question 8

1 / 1 pts

Which SQL query selects those records that have null values for the Address field from the Customer table?

- ☐ SELECT * FROM Customers WHERE Address == NULL;
- ☐ SELECT * FROM Customers WHERE Address == "";
- ☒ SELECT * FROM Customers WHERE Address IS NULL;
- ☐ SELECT * FROM Customers WHERE Address = NULL;



Question 9

1 / 1 pts

What does the SQL statement given below do?

```
SELECT * FROM Customers
WHERE Country IN (SELECT Country FROM Suppliers);
```

- ☐ Selects all the Customers records that have the Country value not null
- ☒ Selects all the Customers records that have the Country value that is present in the Supplier's table
- ☐ Selects all the Suppliers records that have the Country value that is present in the Customers table



PartialQuestion 10

0.67 / 1 pts

Choose all the statements that are NOT true about NoSQL databases

- ☒ All of them follow ACID rules
- ☐ Many of them are implemented with 'eventual consistency' principle
- ☐ Some of them could have implemented a Key/Value store
- ☐ Many of them are highly available systems
- ☒ All of them maintain consistency
- ☐ They all provide ability to build relationship between various entities through primary key/foreign key concepts



IncorrectQuestion 11

0 / 1 pts

Choose the time complexity in Big O notation for the below code chunk

```
z = []
for s in range(n):
    for k in range(m):
        z.append(s * k)
```

- ☒ $O(n^2)$
- ☐ $O(n)$
- ☐ $O(nm)$
- ⋮

Question 12

1 / 1 pts

I have a regex expression as given below

```
r'^([ ^mn])\w{3,5}\1$'
```

What type of line does this regex match?

- ☐ The first letter is m or n, followed by any 3 to 5-word characters and finally, the line ends with 1
- ☐ The first letter is m or n, followed by any 3 to 5-word characters and finally, the line ends with the same character as the first letter.
- ☐ The first letter is not m or n, followed by any 3 to 5-word characters and finally, the line ends with 1
- ☒ The first letter is not m or n, followed by any 3 to 5-word characters and finally, the line ends with the same character as the first letter.

⋮

Question 13

2 / 2 pts

Please take a look at the code below. The last line is the program output from the last statement inside the highlighted section.

Answer two questions:

1. Why is this filter returning all the strings except the empty string?
2. What is the one small change you will make to this code to filter and get only those words that have a string length equal to 5 ('apple' and 'mango' only)? Write the only change you will make as your 2nd answer

```
def get_5_length(x):
    return len(x)

my_list = ['apple', 'mango', '', 'banana', 'kiwi', 'berries']

list(filter(get_5_length, my_list))

['apple', 'mango', 'banana', 'kiwi', 'berries']
```

Your Answer:

1. it is because an empty string will be evaluated as a 'false' in python and the filter will filter that out.
2. return len(x) if len(x) == 5

⋮

Question 14

1 / 1 pts

Consider the snippet of code below

```
d = {1: 'a'}
d[1] = 4
print(d[1])
```

Executing this code will

- ☐ throw an Error, key value '1' is already there in the dictionary
- ☒ print '4' on the console
- ☐ throw an Error, dictionaries do not go by index values
- ☐ throw an Error, original value 'a' is a string and you can only set a string again

⋮

IncorrectQuestion 15

0 / 1 pts

SQLite database starts its own process when it starts and you connect to that process using the Connection object

- ☒ True
- ☐ False

⋮

Question 16

1 / 1 pts

In this course, the Google Cloud solution (the instructions to set up your environment on GCP) of running your JupyterLab is an example of using

- ☒ Virtual Machine and Container

- ☐ Virtual Machine Only
- ☐ Docker Container only
- ☐ Bare metal server



Question 17

1 / 1 pts

When you want to open a text file 'readme.txt' that already has some content in it and you want to add more content to it, you would use which second argument in the **open** statement given below? i.e., ... will be replaced by?

```
f = open("readme.txt", ...)
```

- ☐ 'rt'
- ☐ 'wt'
- ☐ 'ab'
- ☒ 'at'



Question 18

1 / 1 pts

What is the output of the statement given below?

```
my_string = 'Happy Exam Day'
print(f'Wish you a {my_string}')
```

- ☒ Wish you a Happy Exam Day
- ☐ Wish you a {my_string}
- ☐ f'Wish you a Happy Exam Day'
- ☐ f'Wish you a {my_string}'



IncorrectQuestion 19

0 / 1 pts

Some string functions may sometimes modify the existing string; e.g.

```
school = 'umich'
school.upper()
print(school)
```

Output of the print statement would be

UMICH

- ☒ True

☐ False

Quiz Score: 19.97 out of 20

This quiz score has been manually adjusted by +3.799999999999997 points.